

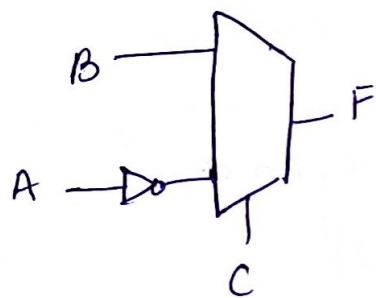
Assignment 3

1. Implement the following function using
 i) 2:1 mux and mux gates.

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

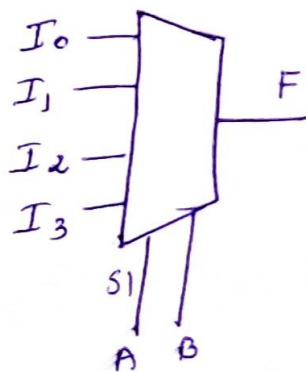
i) 2:1 mux and not

When $C=0$, $F=B$ and $C=1$ $F=\bar{A}$



ii) 4:1 mux

A	B	F
0	0	0
0	1	1
1	0	0
1	1	$\bar{C}$

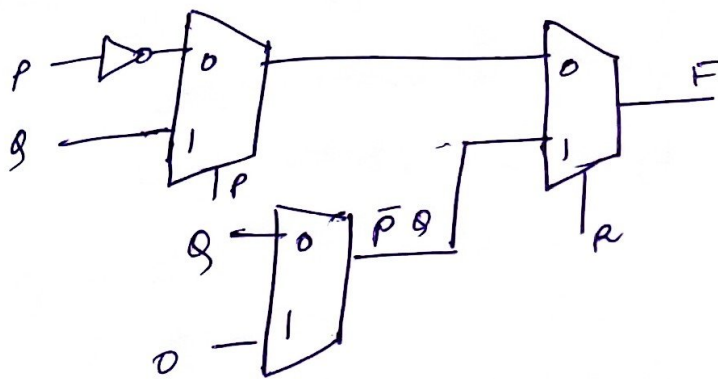


$$b) F(P, Q, R) = \overline{P}\overline{R} + Q\overline{R} + \overline{P}Q$$

i) using 2:1 mux

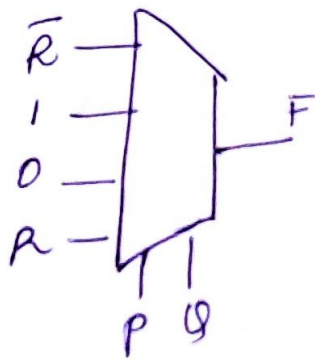
$$\text{when } R=0 \quad F = \overline{P} + Q + \overline{P}Q = \overline{P} + Q$$

$$R=1 \quad F = \overline{P}Q \quad I_0 = \overline{P} + Q \quad I_1 = \overline{P}Q$$



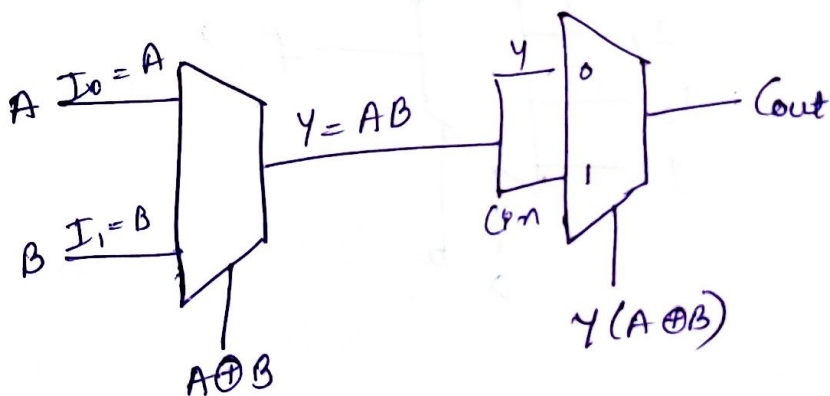
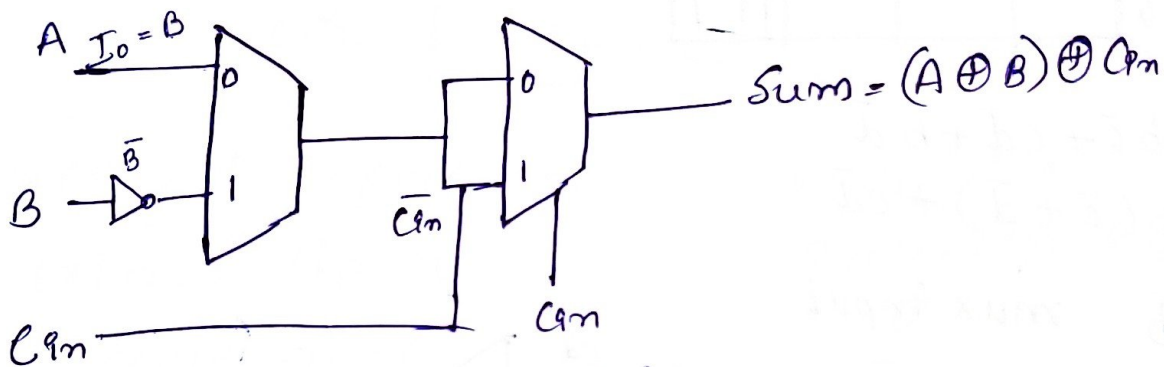
ii) Using 4:1 mux

P	Q	F
0	0	$\overline{R}$
0	1	1
1	0	0
1	1	$\overline{R}$



2. Implement the functionality of full adder using the 2:1 mux only.

Full adder :- $Sum = A \oplus B \oplus C$
 $Carry = AB + C_m(A \oplus B)$



$A \oplus B = \bar{A}B + A\bar{B}$ when $A=1$ o/p = $\bar{B}$
 $A=0$ o/p = B

3. Simplify $f(a,b,c,d) = \sum_m (2,4,5,6,10) + D(12,13,14,15)$
and implement SOP using 4:1 mux

ab \ cd	00	01	11	10
00				1
01	1	1		1
11	X	X		X
10				1

$$f = b\bar{c} + c\bar{d} + b\bar{d}$$

$$f = b(\bar{c} + \bar{d}) + c\bar{d}$$

a	b	mux input
0	0	$I_0 = c\bar{d}$
0	1	$I_1 = 1$
1	0	$I_2 = c\bar{d}$
1	1	$I_3 = 1$

